

OCR Further Mathematics  
Mechanics - Further Dynamics  
Question Paper  
AS and A Level  
Further Mathematics A - H235, H245



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The latest topic questions for the year 2024 / 2025+ exams.  
To be used by all students preparing for OCR AS and A Level Further Mathematics.  
Students of other boards may also find this useful.

- 1(a) A particle  $P$  of mass 3 kg moves on the  $x$ -axis under the action of a single force acting in the positive  $x$ -direction. At time  $t$  s, where  $t \geq 0$ , the displacement of  $P$  is  $x$  m and its velocity is  $v$  ms<sup>-1</sup>. The magnitude of the force acting is inversely proportional to  $(t + 1)^2$ .

Initially  $P$  is at rest at the point where  $x = 1$ . When  $t = 1$ ,  $v = 2$ .

Show that  $\frac{dv}{dt} = \frac{k}{3(t+1)^2}$  where  $k$  is a constant. [1]

- (b) Find an expression for  $v$  in terms of  $t$ . [4]

- (c) Find an expression for  $x$  in terms of  $t$ . [2]

- (d) As  $t$  increases,  $v$  approaches a limiting value,  $v_T$ .

Determine how far  $P$  is from its initial position at the instant when  $v$  is 95% of  $v_T$ . [5]

- 2(a) The resistive force,  $F$ , on a sphere falling through a viscous fluid is thought to depend on the radius of the sphere,  $r$ , the velocity of the sphere,  $v$ , and the viscosity of the fluid,  $\eta$ . You are given that  $\eta$  is measured in N m<sup>-2</sup> s.

By considering its units, find the dimensions of viscosity. [2]

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- (b) A model of the resistive force suggests the following relationship:  $F = 6\pi h^\alpha r^\beta v^\gamma$ .

Explain whether or not it is possible to use dimensional analysis to verify that the constant  $6\pi$  is correct.

[1]

- (c) Use dimensional analysis to find the values of  $\alpha$ ,  $\beta$  and  $\gamma$ .

[4]

- (d) A sphere of radius  $r$  and mass  $m$  falls vertically from rest through the fluid. After a time  $t$  its velocity is  $v$ .

By setting up and solving a differential equation, show that  $e^{-kt} = \frac{g - kv}{g}$  where  $k = \frac{6\pi\eta r}{m}$ .

[5]

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- (e) As the time increases, the velocity of the sphere tends towards a limit called the terminal velocity.

Find, in terms of  $g$  and  $k$ , the terminal velocity of the sphere.

[1]

- (f) In a sequence of experiments the sphere is allowed to fall through fluids of different viscosity, ranging from small to very large, with all other conditions being constant. The terminal velocity of the sphere through each fluid is measured.

Describe how, according to the model, the terminal velocity of the sphere changes as the viscosity of the fluid through which it falls increases.

[2]

- 3(a) A particle  $P$  of mass  $2.4 \text{ kg}$  is moving in a straight line  $OA$  on a horizontal plane.  $P$  is acted on by a force of magnitude  $30 \text{ N}$  in the direction of motion. The distance  $OA$  is  $10 \text{ m}$ .

Find the work done by this force as  $P$  moves from  $O$  to  $A$ .

[2]

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- (b) The motion of  $P$  is resisted by a constant force of magnitude  $R$  N. The velocity of  $P$  increases from  $12 \text{ ms}^{-1}$  at  $O$  to  $18 \text{ ms}^{-1}$  at  $A$ .

Find the value of  $R$ .

[3]



- (c) Find the average power used in overcoming the resistance force on  $P$  as it moves from  $O$  to  $A$ .

[3]

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(d) When  $P$  reaches  $A$  it collides directly with a particle  $Q$  of mass  $1.6 \text{ kg}$  which was at rest at  $A$  before the collision. The impulse exerted on  $Q$  by  $P$  as a result of the collision is  $17.28 \text{ N s}$ .

(i) Find the speed of  $Q$  after the collision.

[2]

(ii) Hence show that the collision is inelastic.

[2]



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4(a) A particle  $P$  of mass  $4.5 \text{ kg}$  is free to move along the  $x$ -axis. In a model of the motion it is assumed that  $P$  is acted on by two forces:

- a constant force of magnitude  $f \text{ N}$  in the positive  $x$  direction;
- a resistance to motion,  $R \text{ N}$ , whose magnitude is proportional to the speed of  $P$ .

At time  $t$  seconds the velocity of  $P$  is  $v \text{ ms}^{-1}$ . When  $t = 0$ ,  $P$  is at the origin  $O$  and is moving in the positive direction with speed  $u \text{ ms}^{-1}$ , and when  $v = 5$ ,  $R = 2$ .

Show that, according to the model,  $\frac{dv}{dt} = \frac{10f - 4v}{45}$ . [2]



(b) (i) By solving the differential equation in the part above, show that  $v = \frac{1}{2}(5f - (5f - 2u)e^{-\frac{4}{45}t})$ . [5]

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(ii) Describe briefly how, according to the model, the speed of  $P$  varies over time in each of the following cases.

- $u < 2.5f$
- $u = 2.5f$
- $u > 2.5f$

[3]



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(c) In the case where  $u = 2f$ , find in terms of  $f$  the exact displacement of  $P$  from  $O$  when  $t = 9$ .

[4]



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- 5 A particle  $P$  of mass 8 kg moves in a straight line on a smooth horizontal plane. At time  $t$  s the displacement of  $P$  from a fixed point  $O$  on the line is  $x$  m and the velocity of  $P$  is  $v$  m s<sup>-1</sup>. Initially,  $P$  is at rest at  $O$ .  
 $P$  is acted on by a horizontal force, directed along the line away from  $O$ , with magnitude proportional to  $\sqrt{9 + v^2}$ . When  $v = 1.25$ , the magnitude of this force is 13 N.

(a) Show that  $\frac{1}{\sqrt{9 + v^2}} \frac{dv}{dt} = \frac{1}{2}$ .

[3]



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(b) Find an expression for  $v$  in terms of  $t$  for  $t \geq 0$ .

[4]



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(c) Find an expression for  $x$  in terms of  $t$  for  $t \geq 0$ .

[2]

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- 6 A particle  $P$  of mass  $m$  moves along the positive  $x$ -axis. When its displacement from the origin  $O$  is  $x$  its velocity is  $v$ , where  $v \geq 0$ . It is subject to two forces: a constant force  $T$  in the positive  $x$  direction, and a resistive force which is proportional to  $v^2$ .

(a) Show that  $v^2 = \frac{1}{k} \left( T - Ae^{-\frac{2kx}{m}} \right)$  where  $A$  and  $k$  are constants.

[5]



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$P$  starts from rest at  $O$ .

- (b) Find an expression for the work done against the resistance to motion as  $P$  moves from  $O$  to the point where  $x = 1$ . [4]



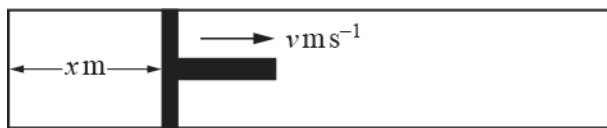
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- (c) Find an expression for the limiting value of the velocity of  $P$  as  $x$  increases. [1]

- 7 A piston of mass 1.5 kg moves in a straight line inside a long straight horizontal cylinder. At time  $t$  s the displacement of the piston from its initial position at one end of the cylinder is  $x$  m and its velocity is  $v$  m s<sup>-1</sup> (see diagram).



The piston starts moving when  $t = 2$  and is brought to rest when it reaches the other end of the cylinder. While the piston is in motion it is acted on by a force of magnitude  $\frac{6}{t^2}$  N in the positive  $x$  direction, and also by a force of magnitude  $\frac{3v}{t}$  N resisting the motion.

- (a) Show that, while the piston is in motion,  $\frac{dv}{dt} + \frac{2v}{t} = \frac{4}{t^2}$ .

[1]

The piston reaches the other end of the cylinder when  $t = 20$ .

(b) Find the speed of the piston immediately before it is brought to rest.

[7]



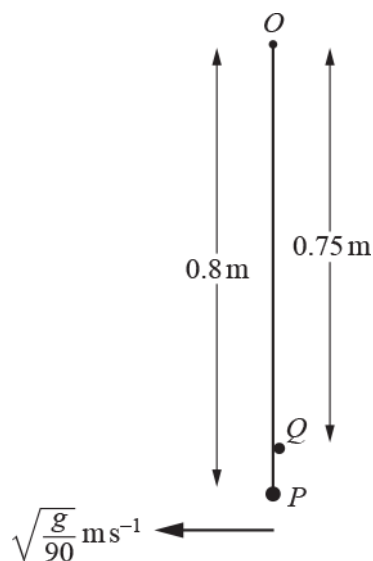
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(c) Show that the piston travels a distance of 5.61 m, correct to 3 significant figures.

[3]



A light inextensible string of length 0.8 m is attached to a fixed point  $O$ . A particle  $P$  of mass  $m$  kg hangs in equilibrium attached to the lower end of the string. A small peg  $Q$  is fixed at a distance 0.75 m vertically below  $O$ .

The particle  $P$  is given a horizontal velocity of  $\sqrt{\frac{g}{90}} \text{ ms}^{-1}$  so that the string initially moves to the left, away from the peg  $Q$  (see diagram).

- (i) Prove that the motion of  $P$  while it is to the left of its initial position is approximately simple harmonic, and find the period of this motion.

[7]



- (ii) Find the time that elapses between the first and second occasions that the string makes an angle of  $5^\circ$  with the vertical. Find also the linear speed of  $P$  in this position. [5]



- (iii) When the string returns to the vertical position,  $P$  begins to move in a circle with centre  $Q$ . Explain with a reason whether the motion of  $P$  is approximately simple harmonic motion while  $P$  is to the right of its initial position. [3]

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- 9 A particle of mass  $0.2 \text{ kg}$  travels in a straight line on a smooth horizontal surface. At time  $t$  seconds it is  $x \text{ m}$  from a fixed point  $O$  and is moving away from  $O$  with velocity  $v \text{ m s}^{-1}$ . A force of magnitude  $\frac{1}{2}\left(12 - \frac{1}{4}v\right)^{\frac{1}{2}} \text{ N}$  acts on the particle in the direction of motion. At time  $t = 0$  the particle is at  $O$  and has velocity  $12 \text{ m s}^{-1}$ .

(i) State the maximum possible velocity of the particle. [1]

(ii) Find an expression for  $v$  in terms of  $t$ , valid while the particle is accelerating. [6]



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(iii) Hence find the distance travelled by the particle as its velocity increases from  $12 \text{ m s}^{-1}$  to  $32 \text{ m s}^{-1}$ . [4]

- 10 A particle  $P$  of mass  $m$  kg is projected vertically upwards through a liquid. Student  $A$  measures  $P$ 's initial speed as  $(8.5 \pm 0.25) \text{ m s}^{-1}$  and they also record the time for  $P$  to attain its greatest height above the initial point of projection as over 3 seconds.

In an attempt to model the motion of  $P$  student  $B$  determines that  $t$  seconds after projection the only forces acting on  $P$  are its weight and the resistance from the liquid. Student  $B$  models the resistance from the liquid to be of magnitude  $mv^2 - 6mv$ , where  $v$  is the speed of the particle.

(a)

(i) Show that  $\frac{dt}{dv} = -\frac{1}{(v-3)^2 + 0.8}$ . [3]

- (ii) Determine whether student  $B$ 's model is consistent with the time recorded by  $A$  for  $P$  to attain its greatest height. [5]

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After attaining its greatest height  $P$  now falls through the liquid. Student  $C$  claims that the time taken for  $P$  to achieve a speed of  $1 \text{ m s}^{-1}$  when falling through the liquid is given by

$$-\int_0^1 \frac{1}{(v-3)^2 + 0.8} dv.$$

- (b) Explain why student  $C$ 's claim is incorrect and write down the integral which would give the correct time for  $P$  to achieve a speed of  $1 \text{ m s}^{-1}$  when falling through the liquid. [3]



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- 11 A body,  $P$ , of mass 2 kg moves under the action of a single force  $F$  N. At time  $t$  s, the velocity of the body is  $v$  m s<sup>-1</sup>, where

$$\mathbf{v} = (t^2 - 3)\mathbf{i} + \frac{5}{2t+1}\mathbf{j} \text{ for } t \geq 2.$$

(a) Obtain  $F$  in terms of  $t$ .

[3]

(b) Calculate the rate at which the force  $F$  is working at  $t = 4$ .

[3]

(c) By considering the change in kinetic energy of  $P$ , calculate the work done by the force  $F$  during the time interval  $2 \leq t \leq 4$ .

[3]

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- 12 A particle  $Q$  of mass  $0.2 \text{ kg}$  is projected horizontally with velocity  $4 \text{ m s}^{-1}$  from a fixed point  $A$  on a smooth horizontal surface. At time  $t \text{ s}$  after projection  $Q$  is  $x \text{ m}$  from  $A$  and is moving away from  $A$  with velocity  $v \text{ m s}^{-1}$ . There is a force of  $3 \cos 2t \text{ N}$  acting on  $Q$  in the positive  $x$ -direction.

- (i) Find an expression for the velocity of  $Q$  at time  $t$ . State the maximum and minimum values of the velocity of  $Q$  as  $t$  varies.

[4]

- (ii) Find the average velocity of  $Q$  between times  $t = \pi$  and  $t = \frac{3}{2}\pi$ .

[4]



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- 13 A particle of mass  $0.4 \text{ kg}$ , moving on a smooth horizontal surface, passes through a point  $O$  with velocity  $10\text{ms}^{-1}$ . At time  $t \text{ s}$  after the particle passes through  $O$ , the particle has a displacement  $x \text{ m}$  from  $O$ , has a velocity  $v\text{ms}^{-1}$  away from  $O$ , and is acted on by a force of magnitude  $\frac{1}{8}v \text{ N}$  acting towards  $O$ . Find
- (i) the time taken for the velocity of the particle to reduce from  $10\text{ms}^{-1}$  to  $5\text{ms}^{-1}$ ,



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[5]

(ii) the average velocity of the particle over this time.



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[6]

- 14 At time  $t = 0$  s a particle  $P$ , of mass  $0.3$  kg, is  $1$  m away from a point  $O$  on a smooth horizontal plane and is moving away from  $O$  with speed  $\sqrt{5} \text{ m s}^{-1}$ . The only horizontal force acting on  $P$  has magnitude  $1.5x$  N, where  $x$  is the distance  $OP$ , and acts away from  $O$ .

- (i) Show that the speed of  $P$ ,  $v$  m s, is given by  $v = \sqrt{5}x$



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[4]

(ii) Find an expression for  $v$  in terms of  $t$ .

[4]



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- 15 A particle  $P$  starts from rest from a point  $A$  and moves in a straight line with simple harmonic motion about a point  $O$ . At time  $t$  seconds after the motion starts the displacement of  $P$  from  $O$  is  $x$  m towards  $A$ . The particle  $P$  is next at rest when  $t = 0.25\pi$  having travelled a distance of 1.2 m.

(i) Find the maximum velocity of  $P$ .



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[3]

(ii) Find the value of  $x$  and the velocity of  $P$  when  $t = 0.7$ .



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[4]

- (iii) Find the other values of  $t$ , for  $0 < t < 1$ , at which  $P$ 's speed is the same as when  $t = 0.7$ . Find also the corresponding values of  $x$ .



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[4]

- 16 A simple pendulum consists of a light inextensible string of length 0.8 m and a particle  $P$  of mass  $m$  kg. The pendulum is hanging vertically at rest from a fixed point  $O$  when  $P$  is given a horizontal velocity of  $0.3 \text{ m s}^{-1}$ .
- (i) Show that, in the subsequent motion, the maximum angle between the string and the downward vertical is 0.107 radians, correct to 3 significant figures.



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[3]

(ii) Show that the motion may be modelled as simple harmonic motion, and find the period of this motion.

[5]

(iii) Find the time after the start of the motion when the velocity of the particle is first  $-0.2 \text{ m s}^{-1}$  and find the angular displacement of  $OP$  from the downward vertical at this time.

[6]



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- 17 A particle  $P$  of mass  $0.4 \text{ kg}$  is projected horizontally with speed  $2 \text{ m s}^{-1}$  from a fixed point  $O$  on a smooth horizontal surface. At times  $t$  s after projection  $P$  is  $x \text{ m}$  from  $O$  and is moving away from  $O$  with speed  $v \text{ m s}^{-1}$ . There is a force of magnitude  $1.6v^2 \text{ N}$  resisting the motion of  $P$ .

- (i) Find an expression for  $\frac{dv}{dx}$  in terms of  $v$ , and hence show that  $v = 2e^{-4x}$ .



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[5]

- (ii) Find the distance travelled by  $P$  in the 0.5 seconds after it leaves  $O$ .



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[5]

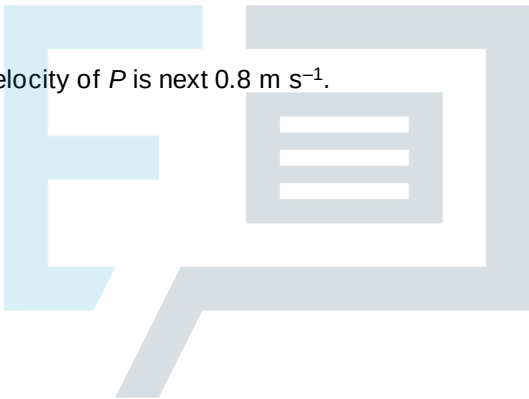
- 18 A particle  $P$  of mass  $0.2 \text{ kg}$  moves on a smooth horizontal plane. Initially it is projected with velocity  $0.8 \text{ m s}^{-1}$  from a fixed point  $O$  towards another fixed point  $A$ . At time  $t \text{ s}$  after projection,  $P$  is  $x \text{ m}$  from  $O$  and is moving with velocity  $v \text{ m s}^{-1}$ , with the direction  $OA$  being positive. A force of  $(1.5t - 1) \text{ N}$  acts on  $P$  in the direction parallel to  $OA$ .

(i) Find an expression for  $v$  in terms of  $t$ .

[3]

(ii) Find the time when the velocity of  $P$  is next  $0.8 \text{ m s}^{-1}$ .

[2]



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(iii) Find the times when  $P$  subsequently passes through  $O$ .



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[4]

(iv) Find the distance  $P$  travels in the third second of its motion.

[2]



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- 19 A particle  $P$  of mass  $m$  kg is attached to one end of a light elastic string of natural length  $0.8$  m and modulus of elasticity  $39.2m$  N. The other end of the string is attached to a fixed point  $O$ . The particle is released from rest at  $O$ .
- (i) Show that, while the string is in tension, the particle performs simple harmonic motion about a point  $1$  m below  $O$ .



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[3]

(ii) Show that when  $P$  is at its lowest point the extension of the string is 0.8 m.



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[3]

(iii) Find the time after its release that  $P$  first reaches its lowest point.



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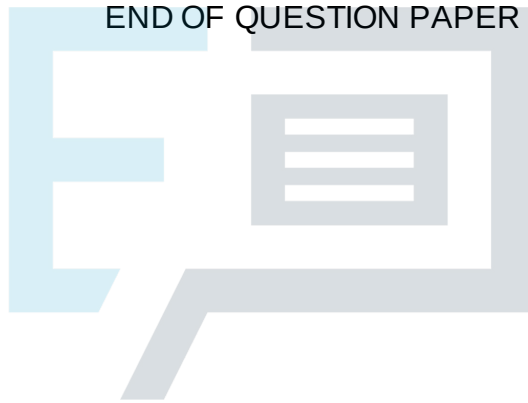
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[6]

(iv) Find the velocity of  $P$  0.8 s after it is released from  $O$ .

[4]

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